Ira A. Fulton Schools of Engineering
Arizona State University

Fall 2026/Spring 2027 Interdisciplinary Capstone Project Request Form

Partner Contact		Project Contact/Mentor	
Company	Intel	Company	Intel
Name	John Rudolph	Name	John Rudolph
Email	john.w.rudolph@intel.com	Email	john.w.rudolph@intel.com
Phone	(480)225-0899	Phone	(480)225-0899

Project Title

Robotic Handoff Characterization and Analysis Tools for Modular Process Equipment in Semiconductor Factory

Project Summary (See page three for instructions)

- Intel is a member of an industry consortium that is defining and developing concepts and standards around a modular, cell-based system for assembly microprocessor units.
- Within this system there are multiple robotic handoffs of material between two separate pieces of equipment. This requires a sophisticated protocol to ensure safe transfer of material and control of independent robots.
- The signal transfer mechanism is across a network connection between the two pieces of equipment. The signals shall be a set of read/write commands that will manage the movement of the robots.
- This project requires a full characterization of the robotic handoff standard/proposal. This will include the full characterization in terms of timing, safety, tolerances, positional teaching and interoperability of systems.
- Additionally, the project will develop a test methodology and tools to monitor and ensure the operation of the handoff protocol signals between the equipment sets.

- Key Concepts:
  - Definition of capability and the key elements of the standards for the interface.
  - Key considerations and challenge areas for the robotic handoff including timing and signal integrity concerns.
  - Prototyping of handoff system including a test system for signal and feedback monitoring.

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Engineering Disciplines Required	None	Low	Med	Hi	Engineering Disciplines Required	None	Low	Med	Hi
Mechanical	☐	☐	☒	☐	Aerospace	☒	☐	☐	☐
Electrical	☐	☐	☒	☐	Computer Science	☐	☐	☒	☐
Industrial	☒	☐	☐	☐	Other	☐	☐	☐	☐

Specialized Skills Required	Low	Med	Hi
	☐	☐	☐
	☐	☐	☐
	☐	☐	☐
	☐	☐	☐
	☐	☐	☐
	☐	☐	☐
	☐	☐	☐

US Citizen ONLY ☐

IP Agreement or NDA Required ☒

Travel Required ☐

Additional Comments:

The development of the standards for this interface has already started within the SEMI organization and the consortia. The implementation has not been done and there are many challenges in the full implementation of the system.

The team will get to work directly with the standards team at Intel and interact through the entire project. The development of the test system will be directly with the Intel team that is working on the concepts.

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Project Title

Semiconductor Process Equipment Energy/Resource Consumption Analysis and Data Collection Process Development

Project Summary (See page three for instructions)

- In the semiconductor industry there is a high focus on keeping the energy and resource consumption for the factories low.
- Intel is working on using data-based approaches to driving specific systems to consume less generation over generation.
- To accomplish this, there is focus on quantifying the current state and then applying techniques and design methods to drive this lower for 'next generation' production equipment.
- This requires understanding of the current consumption and then applying different production techniques or equipment capabilities to drive this consumption to a lower level.
- The project will focus on quantifying current consumption levels, developing tools to measure the current rate of consumption, defining and developing techniques of reducing the consumption and defining procedures and methods for continuously monitoring the levels.

- Key Concepts:
  - Definition of processes and considerations for the establishment of the baseline.
  - Approaches for reduction in consumption for areas of concern in revised system
  - Measurement system for baseline and end results to demonstrate efficiency gains

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Electrical	☐	☐	☒	☐	Computer Science	☐	☒	☐	☐
Industrial	☐	☐	☒	☐	Other	☐	☐	☐	☐

Specialized Skills Required	Low	Med	Hi
	☐	☐	☐
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Additional Comments:

This will entail gaining knowledge of Intel equipment as well as a possibility of seeing the concepts firsthand in the Intel factories.

The expectation for this program is to analyze, measure and define directions that can be put into use to help reduce our consumption.

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Project Title

Automated Delivery of Material/Parts to Process Tools Including Machine Integration for Full Automated Replenishment

Project Summary (See page three for instructions)

- As part of the movement to increased automated operations in the backend assembly processes for semiconductors, there is a need to automate typical human operations for replenishment and delivery of items to process tools.
- To enable this the robotic systems, controls architecture and the mechanical aspects of the delivery need to be developed and standardized to allow for automated delivery.
- The concept is to utilize Autonomous Mobile Robots with integrated secondary robotics to perform these tasks.
- This project scope is to define the operational model, review and develop concepts around the delivery methods, define the controls aspects for managing the delivery and design delivery points and container concepts that can be standardized for this function.

- Key Concepts:
  - Definition of material scope and proposed ‘carrier’ designs.
  - Integration aspects for mechanical delivery and the enabling to the autonomous robots.
  - Specific standardization of the payloads and delivery points for the Process tools.

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Electrical	☐	☐	☒	☐	Computer Science	☐	☐	☒	☐
Industrial	☐	☐	☒	☐	Other	☐	☐	☐	☐

Specialized Skills Required	Low	Med	Hi
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IP Agreement or NDA Required ☒

Travel Required ☐

Additional Comments:

There will be opportunity for prototyping and concept demonstration for the containers and the delivery concepts. There is a possibility for work directly with AMRs to demonstrate the capabilities, but that will need to be finalized.

The expectations are to define the overall ecosystem with targeted demonstrations for the management and delivery of the materials.

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Fall 26/Spring 27 ***Capstone proposals should be received by Friday, July 24th, 2026***

Senior Design Project Proposal Request Form – Instructions

1. The focus of Fulton Senior Design Projects is "brainstorm, plan, design, make, test, [REPEAT] and document". Please create your proposal with this flow in mind.

   Please fill in the sections as noted.

   a. **Project Title:** A short, illustrative project name

   b. **Project Summary:** A summary of the project to familiarize students with the project and get them excited about participating. This may include strategic intent, problem statement, scope, expectations. *The Summary should be suitable for general distribution and should not include any company proprietary information. We recommend that additional documentation is attached, or links included, to provide students with a more thorough understanding of the company, problem statement, and project.*

   c. **Engineering Disciplines Required:** Check the expected competency/experience level for each of the indicated engineering disciplines.

   d. **Specialized Skills Desired –** Indicate any desired specialized skills for the project. If possible/available, we the Capstone Programs will work to assign students with those skills.

   e. **Special Requirements:** Please indicate any restrictions on student participation, IP, or security requirements. We ask that you restrict participation to US Citizens only if you have compelling national security or ITAR restrictions. Any NDAs and/or IP Agreements are executed directly between the project sponsor and the students. *If additional forms or paperwork is required, please attach forms with your application. NDAs, IP agreements, special access requirements, usage of hardware, software licenses, or other special requirements should be addressed ahead of time to avoid delays after project selection*

2. **Fees**
   For the Fall 26/Spring 27 academic year, project fees are required to support the program and projects. Standard Interdisciplinary Capstone Project fee is $7500, with approximately $4000 going directly to the project budget. The Capstone Project fee is based upon the type and size of the Sponsor. Large companies are typically at the $7500 fee level. Smaller organizations, not-for-profit, and Government are typically charged a lower fee. The budget assigned to the team is proportional to the fee. Fee amount will be agreed to prior to acceptance of the project. Partners will be invoiced within 14-24 days after the project selection deadline on September 4. Payment is due 30 days after receipt of the invoicing unless approved otherwise.

   The fee will cover the Capstone Project budget (~55% of fee), including materials and consumables. The remainder directly funds the overall Capstone program labs and resources. Projects requiring significant funds may require additional corporate support. The budget may be discussed prior to the project proposal submission.

If you have more than one concept to propose, please submit each idea on a separate form**.** For more information on the courses and project expectations, please see the Fulton Capstones website: https://capstones.engineering.asu.edu/